

Portland Cement

Categories: [Ceramic](#); [Concrete](#)

Material Notes: Portland cement (Type I) paste with no aggregate prepared with a water/cement ratio of 0.35:1 and cured 28 days.

This data comes from a study that examined the effect of silica fume on Portland cement pastes. A total of four data sheets are in MatWeb from this study.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	2.01 g/cc	0.0726 lb/in ³	
Porosity	2.32 %	2.32 %	Air Void Content

Mechanical Properties	Metric	English	Comments
Tensile Strength, Ultimate	0.910 MPa	132 psi	
Elongation at Break	0.0041 %	0.0041 %	In Tension
Tensile Modulus	11.2 GPa	1620 ksi	
Compressive Strain at Break	1.72 %	1.72 %	
Compressive Strength	57.9 MPa	8400 psi	
Compressive Modulus	2.92 GPa	424 ksi	

Thermal Properties	Metric	English	Comments
Specific Heat Capacity	0.736 J/g-°C	0.176 BTU/lb-°F	
Thermal Conductivity	0.530 W/m-K	3.68 BTU-in/hr-ft ² -°F	
Shrinkage	0.0498 %	0.0498 %	Upon drying

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.